

Learning to Trade

Applying RL into Financial Markets

PPO with Technical & News-Based Sentiment on AAPL (2018–2020)

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Problems

- Can RL agents successfully learn effective trading strategies?
- Where does RL succeed and fail?
- Does incorporating Sentiment signals improve trading decisions?

Hypothesis?

- Expect PPO to achieve similar or slightly better risk-adjusted returns than buy-and-hold in 2020.
- Expect adding sentiment to modestly improve results
- RL will suffer from overfitting issues

Literature review: Reinforcement Learning in Finance

"Direct Reinforcement" bypasses the need for price forecasting. Agents optimize trading positions (buy/sell/hold) directly to maximize the Sharpe ratio / accumulated returns.

(Moody & Saffell, 2001; Deng et al., 2017)

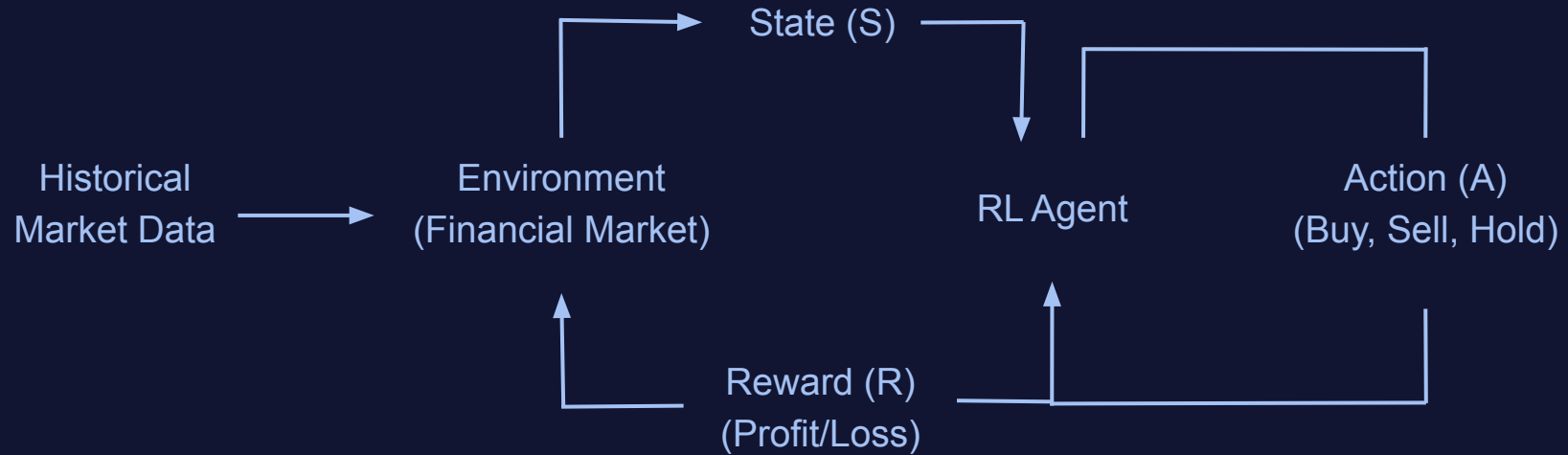
Reinforcement Learning frameworks have been successfully extended to multi-asset environments, using algorithms like Deterministic Policy Gradient to dynamically rebalance portfolio weights.

(Jiang et al., 2017; Kaur, 2017)

Recent surveys highlight the evolution from traditional quantitative methods to Deep Reinforcement Learning, identifying model stability and risk adjusted returns as the primary metrics for modern financial agents.

(Pippas et al., 2025; Bai et al., 2024; Hambly et al., 2023)

Methodology - training process



Environment and Pipeline

- Data: AAPL daily OHLCV (2018 - 2020)
- Features:
 - RSI(14), MA20
 - Calendar: weekday
 - News sentiment: daily aggregated FinBERT scores
 - a. raw vs cleaner variants
- Environment:
 - log-return reward
 - continuous action $[-1, 1]$
 - transaction costs
- Agent: PPO (SB3)
- Evaluation:
 - 2020 (no overlap with training)
 - Metrics: final value, total return, annualized Sharpe (log-returns)

$$\bar{r} = \frac{1}{N} \sum_{t=1}^N r_t^{(p)}, \quad \sigma_r = \sqrt{\frac{1}{N-1} \sum_{t=1}^N (r_t^{(p)} - \bar{r})^2}$$

$$r_t^{(p)} = \log V_t - \log V_{t-1} \quad \text{MA20}_t = \frac{1}{20} \sum_{i=0}^{19} P_{t-i}$$

$$h_t = h_{t-1} + \Delta h_t,$$

$$C_t = C_{t-1} - \Delta h_t P_t - \underbrace{c |\Delta h_t| P_t}_{\text{transaction cost}}.$$

$$V_t^{\text{BH}} = V_0 \cdot \frac{P_t}{P_0} \quad R_t = \alpha \cdot \log \frac{V_t}{V_{t-1}}$$

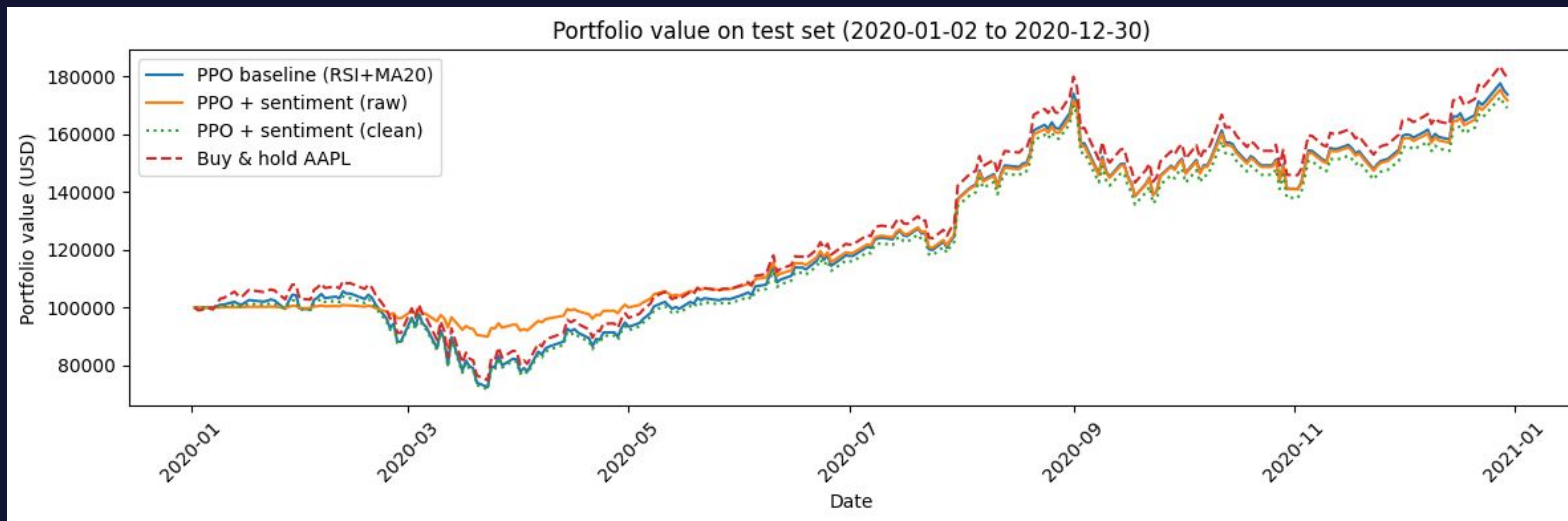
$$\text{TotalReturn} = \left(\frac{V_T}{V_0} - 1 \right) \times 100\%$$

$$\text{Sharpe} = \sqrt{D} \cdot \frac{\bar{r}}{\sigma_r}$$

Problem: Look Ahead Bias

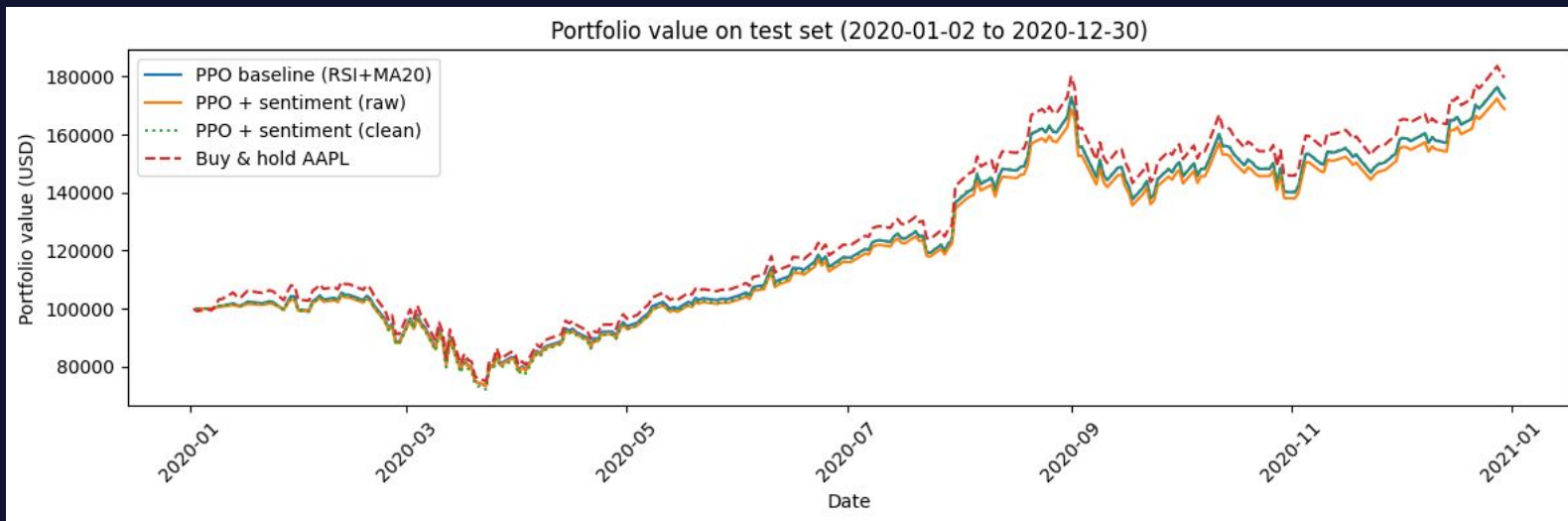
- Strict time split
- Indicators only use information up to time t
- FNSPID news is filtered to the same date range and merged by date
- Remaining Risks:
 - Single asset
 - hyperparameter tuning risks overfitting

Performance - First Run



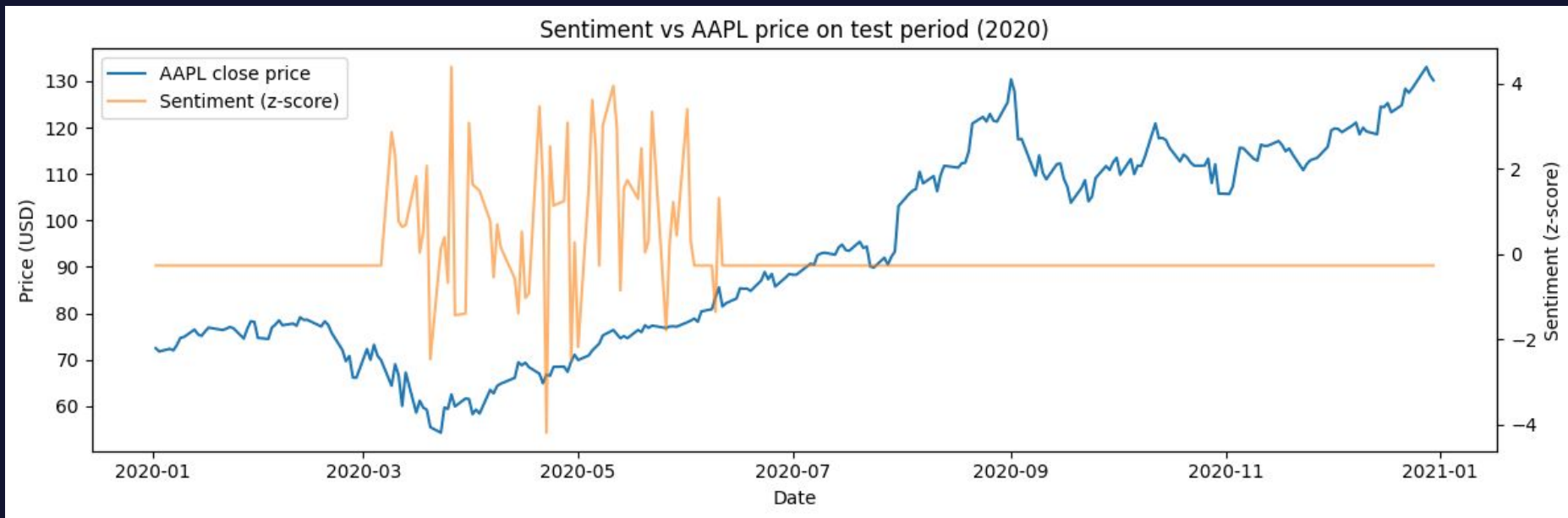
	Strategy	Final value (\$)	Total return (%)	Sharpe (ann., log-returns)
0	PPO baseline (RSI+MA20)	173777.16	73.78	1.192
1	PPO + sentiment (RSI+MA20+sentiment)	171787.66	71.79	1.771
2	PPO + CLEAN sentiment (RSI+MA20+sentiment_clean)	169106.44	69.11	1.161
3	Buy & hold AAPL	179623.50	79.62	1.259

Performance - Second Run



	Strategy	Final value (\$)	Total return (%)	Sharpe (ann., log-returns)
0	PPO baseline (RSI+MA20)	172512.04	72.51	1.202
1	PPO + sentiment (RSI+MA20+sentiment)	168743.52	68.74	1.195
2	PPO + CLEAN sentiment (RSI+MA20+sentiment_clean)	172681.69	72.68	1.186
3	Buy & hold AAPL	179623.52	79.62	1.259

Performance - Sentiment vs Price

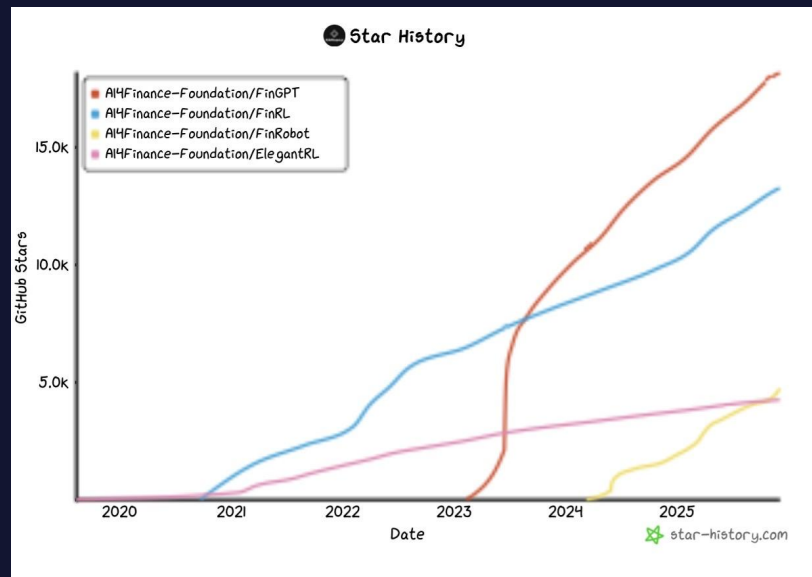


Conclusion

- Single-stock PPO trading framework on APPL with technical and news-based sentiment and FinBert
- PPO variants tracked buy-and-hold performance, but did not reliably outperform it
- Adding daily sentiment produce marginal differences
- Reproducible pipeline for future use

Possible Extensions / Future Work

- GPT4-mini or FinGPT(with llama 2-7b-hf)
- Article title vs Content
- Additional RL Algorithms: DQN, SAC
 - multi-asset portfolio environment
- Different Time Frames and Seeds (with average and standard deviation)



Thank You!